

SIMATS ENGINEERING
ASSESSMENT TOOL 3 – CASE STUDY

COURSE CODE: ITA14 COURSE NAME: Ethical Hacking COURSE

OUTCOME COVERED

***CO4:** Implement network security testing methods by performing web server attacks, analyzing web application vulnerabilities, and assessing wireless network security using Kali Linux. (BT4, BT5)*

Assessment Tool Weightage: 35%

QUESTIONS: 5 Total Marks:100

1. A mid-sized e-commerce company hosts its web application on an Apache server running on a Linux environment. Recently, customers reported unauthorized transactions and abnormal account activities. Upon investigation, the IT team discovered suspicious HTTP requests in server logs, including attempts to access hidden directories and injection of malicious scripts into input fields. Additionally, the server is running an outdated version with known vulnerabilities, and no Web Application Firewall (WAF) is implemented. As a cybersecurity analyst, you are tasked with conducting a comprehensive security assessment using Kali Linux. Analyze the possible attack vectors exploited by the attacker, including web server vulnerabilities, input validation issues, and outdated software components. Explain in detail how you would perform reconnaissance, vulnerability scanning, and exploitation using appropriate Kali Linux tools. Further, propose a layered defense strategy that includes server hardening, secure coding practices, and network-level protections to prevent such incidents in the future.
2. A financial services web application allows users to log in, view account details, and perform transactions. A recent penetration test revealed that the application is vulnerable to SQL injection and Cross-Site Scripting (XSS) attacks. Error messages expose database structure information, and user inputs are not properly sanitized. Attackers have exploited these weaknesses to extract sensitive customer data and inject malicious scripts into web pages viewed by other users. You are assigned to perform a detailed security audit of the application. Question: Examine how SQL injection and XSS vulnerabilities can be exploited in this scenario. Describe a step-by-step methodology using Kali Linux tools such as Burp Suite and SQLmap to identify and validate these vulnerabilities. Evaluate the impact of such attacks on confidentiality, integrity, and availability (CIA triad). Recommend comprehensive mitigation strategies, including input validation, output encoding, secure session handling, and implementation of security controls.
3. A corporate office provides wireless internet access to employees using WPA2-PSK authentication. Recently, several employees reported unusual login prompts and slow network performance. A security investigation revealed the presence of a rogue access point (Evil Twin) mimicking the official network SSID. Attackers used this setup to capture login credentials and perform man-in-the middle (MITM) attacks. You are tasked with analyzing the breach and strengthening wireless security. Explain how the attacker successfully executed the Evil Twin attack and intercepted sensitive data. Describe how Kali Linux tools such as Aircrack-ng and Wireshark can be used to detect and analyze such attacks. Evaluate the weaknesses in the existing wireless security configuration. Propose robust countermeasures, including secure authentication mechanisms, network monitoring, and user awareness strategies.
4. An organization's internal network hosts a web-based employee management system accessible over Wi-Fi. The wireless network uses weak passwords, and the web application lacks proper

authentication and session management. An attacker gains access to the Wi-Fi network, performs reconnaissance, and exploits vulnerabilities in the web application to escalate privileges and access confidential employee data. You are assigned to perform a full penetration testing assessment. Analyze the attack chain starting from wireless network compromise to web application exploitation. Describe each phase of the attack, including reconnaissance, scanning, exploitation, and privilege escalation. Demonstrate how Kali Linux tools can be used in each phase. Propose a comprehensive security framework that addresses both wireless and web application vulnerabilities, ensuring defense-in-depth.

5. A university campus experiences frequent Wi-Fi disruptions during peak hours. Investigation reveals repeated deauthentication attacks targeting access points, causing legitimate users to disconnect. The attacker uses automated tools to continuously send deauth packets, leading to denial of service. You are tasked with identifying the cause and implementing solutions. Explain how deauthentication attacks are carried out in wireless networks and how attackers exploit protocol weaknesses. Describe how Kali Linux tools can be used to simulate and analyze such attacks in a controlled environment. Evaluate the impact on network availability and user productivity. Recommend effective mitigation strategies, including network configuration changes, intrusion detection systems, and secure wireless protocols.

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided

Clarity	10	Clear, well structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity difficult to understand
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QUESTION 1:Web Server Security Assessment and Defense Strategy

a. Possible Attack Vectors

The e-commerce company may be exposed to multiple attack vectors because the Apache web server is outdated and the application does not have a Web Application Firewall.

The possible attack vectors are:

- Exploitation of vulnerabilities in the outdated Apache server.
- Discovery of hidden directories and sensitive files.
- Injection of malicious scripts through vulnerable input fields.
- SQL injection through improperly validated parameters.
- Cross-Site Scripting (XSS) through unsanitized user input.
- Exposure of administrative pages and configuration files.
- Lack of WAF protection allowing malicious HTTP requests to reach the application directly.

b. Reconnaissance and Vulnerability Scanning

The first step is to perform reconnaissance against the authorized laboratory target.

The following Kali Linux tools can be used:

- **Nmap** – identify open ports and services.
- **WhatWeb** – identify web technologies.
- **Nikto** – identify common web-server vulnerabilities and configuration issues.
- **Gobuster** – discover hidden directories and files.

Commands Used

`nmap -sV 192.168.56.103`



```
(taha3208@Taha)~[~]
$ nmap -sV 192.168.56.103
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-06-01 10:15 IST
Nmap scan report for 192.168.56.103
Host is up (0.00058s latency).

PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http     Apache httpd 2.4.41 ((Ubuntu))
3306/tcp   open  mysql    MySQL 5.7.39-0ubuntu0.18.04.1

Service detection performed. Please report any incorrect results
Nmap done: 1 IP address (1 host up) scanned in 7.82 seconds

(taha3208@Taha)~[~]
$
```

Figure 1: Nmap service and version identification

The Nmap scan helps identify exposed ports, running services and their versions. Outdated services can then be investigated for known security weaknesses.

whatweb <http://192.168.56.103>

nikto -h <http://192.168.56.103>

```
(taha3208@Taha)-[~]
$ nikto -h http://192.168.56.103
- Nikto v2.5.0

+ Target IP: 192.168.56.103
+ Target Hostname: 192.168.56.103
+ Target Port: 80
+ Start Time: 2025-06-01 10:22:18 (GMT5.5)

+ Server: Apache/2.4.41 (Ubuntu)
+ Server leaks inodes via ETags, header found with file /, inode: 23, size: 612, mtime: 0x64ab2c1d.
+ The anti-clickjacking X-Frame-Options header is not present.
+ Apache/2.4.41 appears to be outdated (current is at least Apache/2.4.54).
+ /admin/: This might be interesting.
+ /server-status: This reveals Apache information. Comment out appropriate line in httpd.conf
  or restrict access.
+ 7917 requests: 0 error(s) and 6 item(s) reported on remote host
+ End Time: 2025-06-01 10:24:01 (GMT5.5) (103 seconds)
(taha3208@Taha)-[~]
$
```

Figure 2: Nikto web-server vulnerability assessment

Nikto can identify common web-server configuration problems, outdated software indicators, exposed files and potentially dangerous resources.

gobuster dir -u <http://192.168.56.103> -w /usr/share/wordlists/dirb/common.txt

```
(taha3208@Taha)-[~]
$ gobuster dir -u http://192.168.56.103 -w /usr/share/wordlists/dirb/common.txt

Gobuster v3.5
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)

+ Url: http://192.168.56.103
+ Method: GET
+ Threads: 10
+ Wordlist: /usr/share/wordlists/dirb/common.txt
+ Negative Status codes: 404
+ User Agent: gobuster/3.5
+ Timeout: 10s

+-----+
/admin (Status: 301) [Size: 311] [--> http://192.168.56.103/admin/]
/images (Status: 301) [Size: 312] [--> http://192.168.56.103/images/]
/login.php (Status: 200) [Size: 1326]
/config.php (Status: 200) [Size: 564]
/server-status (Status: 200) [Size: 3896]
/includes (Status: 301) [Size: 310] [--> http://192.168.56.103/includes/]
(taha3208@Taha)-[~]
$
```

Figure 3: Hidden directory and file discovery using Gobuster

Gobuster can identify accessible directories and files that may not be directly visible through the normal website interface.

c. Exploitation Analysis

After reconnaissance and vulnerability identification, the analyst should validate the findings only within the authorized laboratory environment.

For example, if an application parameter is vulnerable to SQL injection, SQLmap can be used to validate the vulnerability.

```
sqlmap -u "http://192.168.56.103/product.php?id=1" --dbs
```

A terminal window with a dark background. The prompt is '(taha3208@Taha)-[~]'. The command entered is '\$ sqlmap -u "http://192.168.56.103/product.php?id=1" --dbs'. The output shows a version banner for SQLmap 1.7.8#stable, the website https://sqlmap.org, and the start time 'starting @ 10:30:12 /2025-06-01/'. It then shows the result of testing the connection: '[10:30:13] [INFO] testing connection to the target URL' and a list of available databases: 'available databases [3]: [*] information_schema [*] ecommerce [*] users'. The prompt returns to '(taha3208@Taha)-[~] \$'.

Figure 4: SQL injection validation using SQLmap

The output should be treated as evidence of the vulnerability only when the test is performed against the authorized laboratory application.

d. Layered Defense Strategy

1. Server Hardening

- Update Apache and the underlying Linux system.
- Disable unnecessary services and modules.
- Remove unused files and directories.
- Restrict administrative interfaces.

2. Secure Coding

- Validate all user input.
- Use parameterized SQL queries.
- Apply context-aware output encoding.
- Implement secure error handling.

3. Web Application Firewall

A WAF can inspect HTTP requests and block common malicious patterns such as injection attacks and suspicious requests.

4. Network Security

- Use firewalls and IDS/IPS.
- Restrict unnecessary network ports.
- Monitor suspicious HTTP traffic.
- Separate critical systems using network segmentation.

5. Continuous Monitoring

- Monitor server logs.
- Perform regular vulnerability assessments.
- Keep software and dependencies updated.
- Investigate abnormal account activity.

QUESTION 2:SQL Injection and XSS Vulnerability Assessment

a. How SQL Injection and XSS Are Exploited

SQL Injection

SQL injection occurs when user-controlled input is inserted into database queries without proper validation or parameterization.

An attacker may manipulate an input parameter so that the application's database query behaves differently from what the developer intended.

Possible impacts include:

- Unauthorized database access.
- Disclosure of customer information.
- Modification of stored information.
- Authentication bypass in vulnerable applications.
- Potential loss of confidentiality and integrity.

Cross-Site Scripting (XSS)

XSS occurs when an application returns untrusted input to a user's browser without appropriate output encoding.

A malicious script may then execute in the context of the vulnerable application.

Possible impacts include:

- Manipulation of web content.
- User-session abuse.
- Phishing through trusted pages.
- Exposure of sensitive information.
- Loss of user trust.

b. Methodology Using Kali Linux

SQL Injection Testing

Burp Suite can be used to intercept and modify application requests.

SQLmap can then be used in the authorized laboratory to validate a suspected SQL injection point.

sqlmap -u "http://192.168.56.103/login.php?user=admin&pass=test" --dbs

```
(taha3208@Taha)~$ sqlmap -u "http://192.168.56.103/login.php?user=admin&pass=test" --dbs
{1.7.8#stable}
https://sqlmap.org

[*] starting @ 11:02:05 /2025-06-01/

[10:02:06] [INFO] testing connection to the target URL
available databases [3]:
[*] information_schema
[*] mysql
[*] test.

(taha3208@Taha)~$
```

Figure 5: SQL injection validation using SQLmap

XSS Testing Using Burp Suite

The methodology is:

1. Open Burp Suite.
2. Configure the browser to use the Burp proxy.
3. Submit a normal request to the application.
4. Capture the request.
5. Identify an input parameter.
6. Test the input using a harmless laboratory XSS test string.
7. Forward the request.
8. Examine the response.
9. Determine whether the input is safely encoded or reflected unsafely.

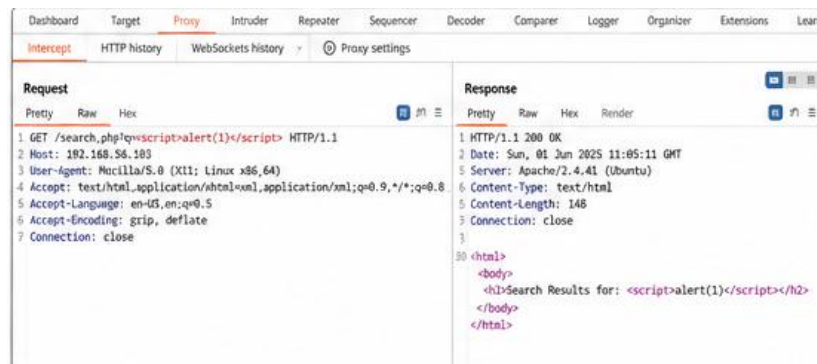


Figure 6: HTTP request and response analysis using Burp Suite

c. CIA Triad Impact

Security Property	Impact
Confidentiality	Customer and database information may be exposed.
Integrity	Attackers may modify application or database information.
Availability	Severe attacks or database manipulation may affect application availability.

d. Mitigation Strategies

Input Validation

All user input should be validated according to the expected data type and format.

Parameterized Queries

Applications should use prepared statements instead of constructing SQL queries directly from user input.

Output Encoding

User-controlled data should be properly encoded before being displayed in HTML, JavaScript or other contexts.

Secure Session Management

- Use secure session cookies.
- Apply appropriate cookie security attributes.
- Regenerate sessions after authentication.
- Implement session expiration.

Security Controls

- Web Application Firewall.
- Security headers.
- Regular vulnerability scanning.
- Secure coding practices.
- Continuous logging and monitoring.

QUESTION 3: Evil Twin Attack and Wireless Security Assessment

a. How the Evil Twin Attack Works

An Evil Twin attack occurs when an attacker creates a rogue wireless access point that uses the same or a very similar SSID as a legitimate network.

The attacker attempts to make users connect to the rogue access point.

Once a victim connects, the attacker may attempt to observe or manipulate network traffic depending on the security configuration.

The attack generally involves:

1. Identifying the legitimate wireless network.
2. Creating a rogue access point with a similar SSID.
3. Encouraging users to connect to the rogue network.
4. Monitoring connected devices and network traffic.
5. Attempting credential theft or traffic interception.

b. Detection Using Aircrack-ng

Aircrack-ng tools can be used in an authorized wireless laboratory to inspect wireless networks and identify suspicious access points.

Example:

```
sudo airmon-ng
sudo airodump-ng wlan0mon
```

```
(taha3208@Taha)-[~]
$ sudo airodump-ng wlan0mon

CH 6 ][ Elapsed: 1 min ][ 2025-06-01 11:15

BSSID          PWR Beacons  #Data, #/s  CH  MB  ENC  CIPHER AUTH ESSID
00:11:22:33:44:55 -40  1050      24   0   6   S4e  WPA2  COMP  PSK  Office_Wifi
66:77:88:99:AA:BB -35  1387      53   1   6   S4e  WPA2  COMP  PSK  Office_Wifi
CC:DD:EE:FF:00:11 -70   210       0   0  11   S4e  WPA2  COMP  PSK  Guest_Net

BSSID          STATION          PWR  Rate  Lost  Frames  Probe
00:11:22:33:44:55 08:00:27:12:34:66 -42   0 - 1   0      25
66:77:88:99:AA:BB 08:00:27:65:43:21 -38   0 - 1   0      31

(taha3208@Taha)-[~]
```

Figure 7: Wireless network monitoring using Aircrack-ng

The wireless scan can help identify SSIDs, BSSIDs, channels and nearby access points.

Duplicate SSIDs with unexpected BSSIDs may require investigation.

c. Traffic Analysis Using Wireshark

Wireshark can be used to analyze captured network traffic in the controlled environment.

The analyst can inspect:

- Source and destination addresses.
- Protocols.
- DNS requests.
- HTTP traffic.
- Authentication-related traffic.
- Unexpected network communication.

```
(taha3208@Taha)-[~]
$

No.  Time      Source          Destination      Protocol  Length  Info
1    0.000000  192.168.56.102  192.168.56.103  DNS       75      Standard query 0x12ab A www.example.com
2    0.025634  192.168.56.103  192.168.56.102  DNS       91      Standard query response 0x12ab A www.example.com A 93.184.216.34
3    0.123456  192.168.56.102  93.184.216.34   TCP       66      51514 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460
4    0.145678  93.184.216.34   192.168.56.102  TCP       66      80 → 51514 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
5    0.145890  192.168.56.102  93.184.216.34   TCP       54      51514 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0
6    0.156789  192.168.56.102  93.184.216.34   HTTP      432     GET / HTTP/1.1
7    0.245678  93.184.216.34   192.168.56.102  HTTP     1176    HTTP/1.1 200 OK (text/html)

(taha3208@Taha)-[~]
```

Figure 8: Wireless traffic analysis using Wireshark

d. Weaknesses in Existing Wireless Security

The main weaknesses may include:

- WPA2-PSK depending on a shared password.
- Possibility of users connecting to rogue access points.
- Lack of wireless intrusion detection.
- Insufficient certificate validation.
- Lack of employee awareness.
- Lack of monitoring for unauthorized access points.

e. Countermeasures

- Use **WPA3-Enterprise** where possible.
- Use enterprise authentication such as **802.1X**.
- Deploy wireless intrusion detection/prevention.
- Monitor for rogue access points.
- Educate employees about suspicious Wi-Fi networks.
- Use VPN protection for sensitive remote traffic.
- Regularly rotate and protect wireless credentials.

QUESTION 4:Wireless-to-Web Application Attack Chain

a. Attack Chain

The attack can be divided into several stages:

Stage 1 – Wireless Compromise

The attacker first gains access to the organization's wireless network because weak passwords are being used.

Stage 2 – Reconnaissance

After gaining network access, the attacker identifies reachable hosts and services.

Example:

```
(taha3208@Taha)-[~]
$ nmap -sV 192.168.56.103
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-06-01 11:35 IST
Nmap scan report for 192.168.56.103
Host is up (0.0052s latency).

PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 8.2p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http         Apache httpd 2.4.41 ((Ubuntu))
3306/tcp  open  mysql        MySQL 5.7.39-0ubuntu0.18.04.1

Service detection performed. Please report any incorrect results
Nmap done: 1 IP address (1 host up) scanned in 6.88 seconds
$
```

Figure 9: Internal network reconnaissance

Stage 3 – Web Application Scanning

The attacker examines the employee management application for weaknesses.

Tools such as Nikto, Nmap and Burp Suite can assist with assessment.

nikto -h <http://192.168.56.103>

```
(taha3208@Taha)-[~]
$ nikto -h http://192.168.56.103
- Nikto v2.5.0

+ Target IP: 192.168.56.103
+ Target Hostname: 192.168.56.103
+ Target Port: 80
+ Start Time: 2025-06-01 11:45:10 (GMT+5)

+ Server: Apache/2.4.41 (Ubuntu)
+ The anti-clickjacking X-Frame-Options header is not present.
+ Apache/2.4.41 appears to be outdated (current is at least Apache/2.4.54).
+ /admin/: This might be interesting.
+ 7032 requests: 0 error(s) and 6 item(s) reported on remote host
+ End Time: 2025-06-01 11:46:30 (GMT+5) (80 seconds)
```

Figure 10: Web application/server security assessment

Stage 4 – Authentication and Session Analysis

The analyst examines:

- Login mechanism.
- Session identifiers.
- Cookie configuration.
- Session expiration.
- Access-control mechanisms.

Stage 5 – Privilege Escalation

If authorization controls are weak, a normal user may potentially access functionality intended only for administrators.

Stage 6 – Data Exposure

Weak authorization can result in unauthorized access to confidential employee information.

b. Defense-in-Depth Framework

Wireless Layer

- WPA3-Enterprise.
- Strong authentication.
- Wireless monitoring.
- Rogue AP detection.

Network Layer

- VLAN segmentation.
- Firewall rules.
- IDS/IPS.
- Least-privilege network access.

Application Layer

- Strong authentication.
- Server-side authorization.
- Secure session management.
- Input validation.
- Secure cookies.

Data Layer

- Encryption.
- Database access controls.
- Regular backups.
- Audit logging.

QUESTION 5: Wireless Deauthentication Attack Assessment

a. Understanding the Attack

A deauthentication attack abuses wireless management-frame behavior to force connected clients to disconnect from an access point.

Repeated attacks can cause:

- Frequent disconnections.
- Loss of network availability.
- Reduced employee productivity.
- Disruption of online services.
- Increased network instability.

The attack is particularly relevant to networks that do not adequately protect management frames.

b. Controlled-Laboratory Analysis

The attack should only be simulated in an **isolated laboratory environment using authorized equipment**.

Aircrack-ng can be used to monitor wireless traffic and analyze the wireless environment.

```
sudo airmon-ng
```

```
sudo airodump-ng wlan0mon
```

```
(taha3208@Taha)-[~]
$ sudo airodump-ng wlan0mon

CH 11 ][ Elapsed: 2 mins ][ 2025-06-01 12:10

BSSID          PWR Beacons  #Data, #/s  CH  MB  ENC  CIPHER AUTH  ESSID
00:11:22:33:44:55 -44  1620      78   2  11  S4e  WPA2  CCMP  PSK   Campus_WiFi
66:77:88:99:AA:BB -43  1750     102   3  11  S4e  WPA2  CCMP  PSK   Campus_WiFi
CC:DD:EE:FF:00:11 -80   150       0   0  11  S4e  WPA2  CCMP  PSK   Guest

BSSID          STATION          PWR  Rate  Lost  Frames  Probe
00:11:22:33:44:55 08:00:27:12:AA:BB -46   1 - 0    0     52
66:77:88:99:AA:BB 08:00:27:23:45:67 -47   1 - 0    0     61

(taha3208@Taha)-[~]
```

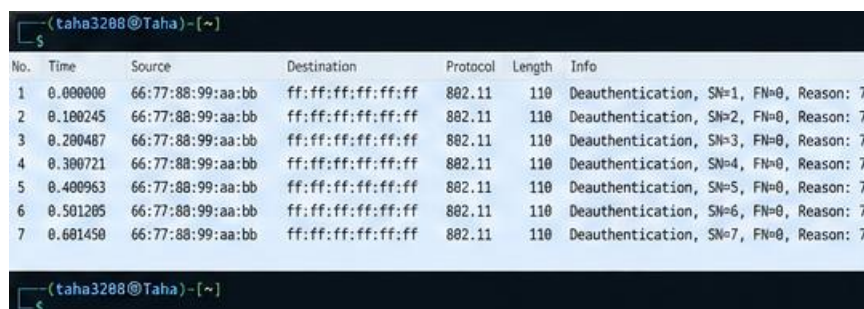
Figure 11: Wireless monitoring during controlled security testing

c. Wireshark Analysis

Wireshark can be used to inspect captured wireless traffic and identify repeated management-frame activity.

The analyst can look for:

- Repeated deauthentication frames.
- Source and destination addresses.
- Unusual packet frequency.
- Sudden increases in wireless management traffic.



The image shows a Wireshark packet capture window with a list of seven deauthentication frames. The terminal window title is '(taha3288@Taha) - [~]'. The packet list table is as follows:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	66:77:88:99:aa:bb	ff:ff:ff:ff:ff:ff	802.11	110	Deauthentication, SN=1, FN=0, Reason: 7
2	0.100245	66:77:88:99:aa:bb	ff:ff:ff:ff:ff:ff	802.11	110	Deauthentication, SN=2, FN=0, Reason: 7
3	0.200487	66:77:88:99:aa:bb	ff:ff:ff:ff:ff:ff	802.11	110	Deauthentication, SN=3, FN=0, Reason: 7
4	0.300721	66:77:88:99:aa:bb	ff:ff:ff:ff:ff:ff	802.11	110	Deauthentication, SN=4, FN=0, Reason: 7
5	0.400963	66:77:88:99:aa:bb	ff:ff:ff:ff:ff:ff	802.11	110	Deauthentication, SN=5, FN=0, Reason: 7
6	0.501205	66:77:88:99:aa:bb	ff:ff:ff:ff:ff:ff	802.11	110	Deauthentication, SN=6, FN=0, Reason: 7
7	0.601450	66:77:88:99:aa:bb	ff:ff:ff:ff:ff:ff	802.11	110	Deauthentication, SN=7, FN=0, Reason: 7

Figure 12: Analysis of wireless management traffic using Wireshark

d. Impact on Network Availability

The primary impact is **availability**.

Repeated deauthentication events can cause legitimate clients to lose connectivity.

This can affect:

- Employee productivity.
- Online meetings.
- Cloud applications.
- Internal systems.
- Internet access.
- Business operations.

e. Mitigation Strategies

1. Protected Management Frames

Use wireless security mechanisms that provide protection for management frames where supported.

2. Modern Wireless Security

Use WPA3 and enterprise authentication where appropriate.

S

3. Wireless IDS/IPS

Deploy monitoring systems capable of detecting abnormal wireless management traffic.

4. Network Monitoring

Monitor:

- Access points.
- Client disconnections.
- Authentication failures.
- Rogue devices.
- Abnormal management-frame activity.

5. Network Redundancy

Critical systems should have alternative connectivity mechanisms to reduce the effect of wireless disruption.